



University of Camerino

SCIENCE AND TECHNOLOGIES

Master Degree in Computer Science (LM-18 Class)
Autonomous and collaborative robotics

Smart City, with ROS2 and Gazebo

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A.Y. 2025/2026

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1. Specification

The objective of this project is to develop a smart city simulation using ROS2 and Gazebo, in which different types of vehicles share the same urban road network and coordinate to provide efficient and safe transportation services.

The simulated environment consists of an urban map organized as a grid of roads and intersections. The city includes bus stops, parking areas, traffic lights, pedestrian crossings, and obstacles distributed throughout the road network. The simulation involves three main categories of vehicles: buses, taxis, and private cars.

Buses follow predefined routes and connect different bus stops across the city. Taxis provide an on-demand transportation service. Passenger requests are generated dynamically during the simulation and include both a pickup location and a destination.

All vehicles are equipped with simulated sensors, including LIDAR and localization systems, which are used to perceive the surrounding environment and estimate their position. Using this information, vehicles must be able to avoid both static and dynamic obstacles, obey traffic signals, and maintain safe navigation even in the presence of traffic.

The presence of obstacles represents one of the main challenges of the simulation. In addition to permanent urban infrastructure, the system introduces road barriers and moving pedestrians that may temporarily obstruct vehicle routes. In such situations, agents must adapt their behavior by slowing down, stopping, or recalculating their paths.

The final objective is to evaluate the effectiveness of a distributed urban mobility system in which multiple autonomous agents share information and resources while simultaneously addressing challenges related to navigation, coordination, traffic management, and collision avoidance.

2. Vehicle State Design

The environment is built upon the integration of ROS2 and Gazebo, which are respectively responsible for the control logic and the physical simulation of the virtual world.

Communication between ROS2 nodes and Gazebo is achieved through the `ros_gz_bridge` package, which converts messages generated by the simulator into standard ROS2 topics. This allows the controllers to receive data from simulated sensors, such as LIDAR scans and localization information, while sending motion commands to the vehicles operating within the environment.

The system is composed of several autonomous agents, each responsible for a specific function within the smart city. Although buses and taxis perform different tasks, their behaviour is modeled through Finite State Machines (FSMs), allowing the various operational phases of the simulation to be managed in a clear and predictable manner.

2.1 Bus Internal States

Buses represent the public transportation service of the city. Each vehicle follows a predefined route composed of a sequence of bus stops distributed throughout the road network.

At the beginning of the simulation, the vehicle is in the *PARKED* state, corresponding to its assigned parking area. After a short waiting period, the bus transitions to the *STARTING_ROUTE* state and begins following its configured route.

During normal operation, the vehicle remains in the *ON_ROUTE* state, where it follows route waypoints while interacting with traffic lights, obstacles, and other road users. Whenever a scheduled stop is reached, the bus temporarily enters the *AT_STOP* state, where it performs a short stop before resuming its journey.

After completing the configured number of route cycles, the bus transitions to the *RETURNING_TO_PARKING* state and navigates back to its assigned parking area. Once the parking maneuver is completed, the vehicle returns to the *PARKED* state and terminates its operational cycle.

2.2 Taxi Internal States

Taxis provide an on-demand transportation service. When they are not serving passengers, they remain in the *IDLE* state, waiting for new transportation requests generated during the simulation.

Upon receiving a new request, the available taxis participate in a distributed coordination procedure to determine which vehicle will be assigned to the service. The

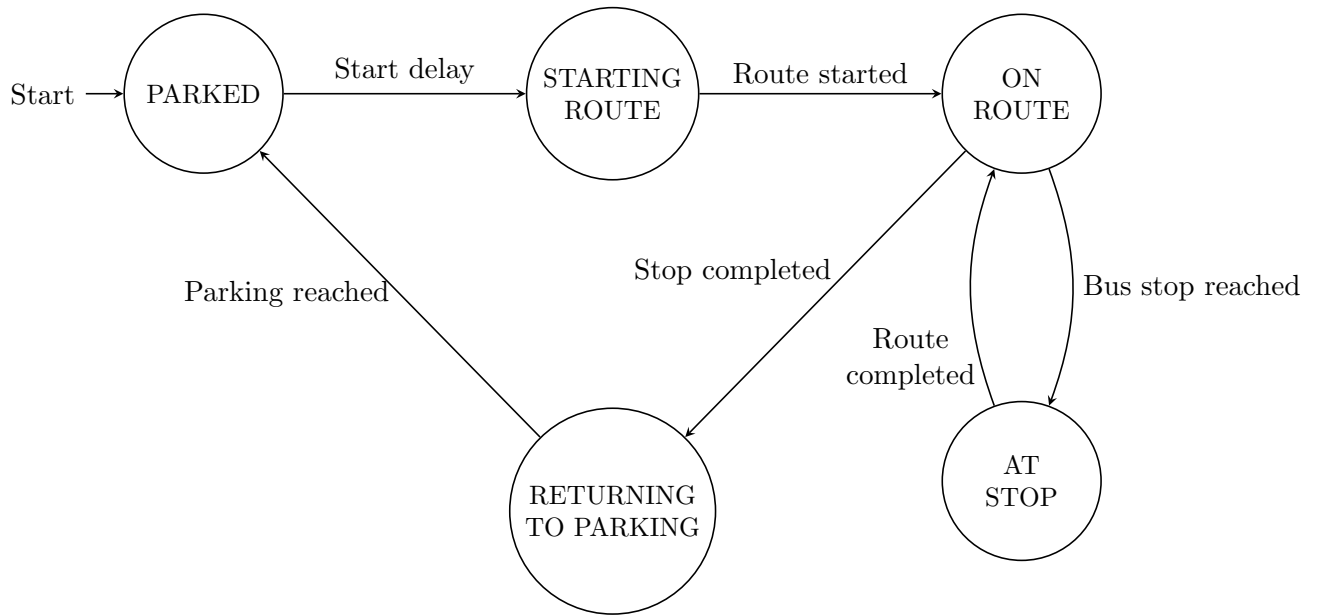


Figure 2.1: Finite State Machine of the Bus

selected taxi then enters the `GOING_TO_PICKUP` state and travels toward the passenger pickup location.

Once the pickup point has been reached, the vehicle transitions to the `TRANSPORTING_PASSENGER` state and navigates toward the final destination. After completing the trip, the taxi enters the `RETURNING_TO_PARKING` state and moves toward an available parking area.

Finally, once parked, the vehicle returns to the `IDLE` state and becomes available for future transportation requests.

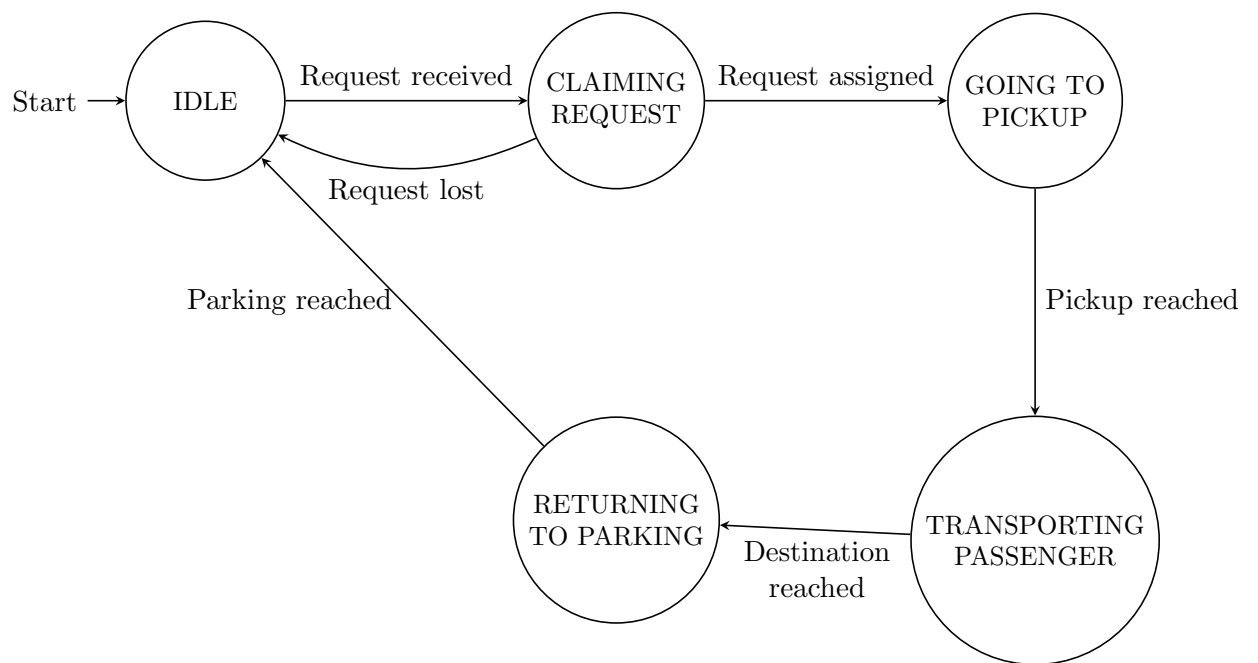


Figure 2.2: Finite State Machine of the Taxi

3. Coordination Algorithm and Communication Architecture

Each vehicle operates autonomously, but it must still share information with the nodes responsible for managing the environment. Nodes publish messages on shared topics, while other interested components receive them through subscriptions. This approach allows the logic of individual agents to remain independent, avoiding direct dependencies between the various modules of the simulation.

3.1 Taxi Coordination

When a new transportation request is published, the available taxis first evaluate their internal state. Only vehicles that are currently free can participate in the assignment process. Each eligible taxi enters a claiming phase, during which it communicates its availability to the other taxis.

After a short waiting interval, the taxis compare the received claims and determine which vehicle should be assigned to the request. The selected taxi then publishes an assignment message, while all other vehicles ignore the request and remain available for future services.

This mechanism prevents multiple taxis from responding to the same request and allows transportation tasks to be distributed among the available vehicles without introducing a single centralized node responsible for all decisions.

3.2 Communication Between Components

The simulation uses several ROS2 topics to connect the main system modules.

Taxis receive transportation requests through a dedicated request topic and communicate with each other using topics dedicated to claims and assignments. In this way, each request is first made visible to all taxis and is then assigned to a single vehicle.

Buses publish their operational status on a dedicated topic, allowing the system to monitor whether they are currently following their route, waiting at a bus stop, or returning to their parking area.

Traffic lights periodically publish their current state, indicating which axis of an intersection is allowed to proceed. Approaching vehicles receive this information and decide whether to continue, slow down, or stop before entering the intersection. Additionally, vehicles can send priority requests to traffic lights in order to reduce waiting times when necessary.

Overall, this architecture clearly separates the responsibilities of the different nodes:

vehicles handle their own navigation, manager nodes coordinate specific services, and traffic lights regulate road intersections.

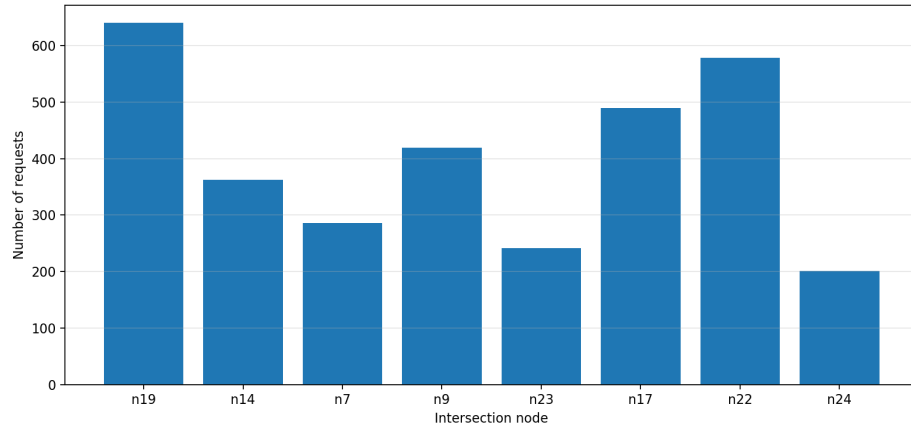


Figure 3.1: Number of traffic priority requests generated at each controlled intersection. The results show which intersections were more frequently involved in vehicle coordination and traffic light priority management.

4. Simulation and Results

Several simulation scenarios were executed within the urban environment developed in Gazebo in order to evaluate the behaviour of the system. During the experiments, buses, taxis, traffic lights, private cars, pedestrian crossings, and road obstacles were simultaneously present within the simulated environment.

The main objective of the tests was to verify the ability of the vehicles to navigate correctly throughout the city while maintaining a coherent behaviour in the presence of traffic and unexpected situations.

4.1 Navigation and Traffic Management

The obtained results show that the vehicles were able to follow their assigned routes while respecting the structure of the road network and maintaining proper interaction with the other agents present in the environment.

Buses successfully completed their routes, performing the expected stops at bus stations and returning to their parking areas at the end of the service. Similarly, private cars reached their selected destinations and generated a continuous traffic flow throughout the simulation.

Taxis correctly completed the transportation requests generated during execution, reaching the pickup locations and transporting passengers to their destinations.

4.2 Obstacle and Pedestrian Crossing Management

A significant portion of the tests was dedicated to evaluating vehicle behaviour in the presence of both static and dynamic obstacles.

The road barriers introduced into the simulation allowed the system to verify its ability to recognize blocked situations and adapt navigation accordingly. In these cases, vehicles slowed down or modified their routes in order to avoid collisions and continue their journey safely.

The presence of pedestrians at road crossings also produced satisfactory results. Vehicles were able to detect occupied road sections and reduce their speed or temporarily stop until safe conditions were restored.

4.3 Distributed Coordination

The experiments also highlighted the correct operation of the distributed coordination mechanism adopted by the taxi service.

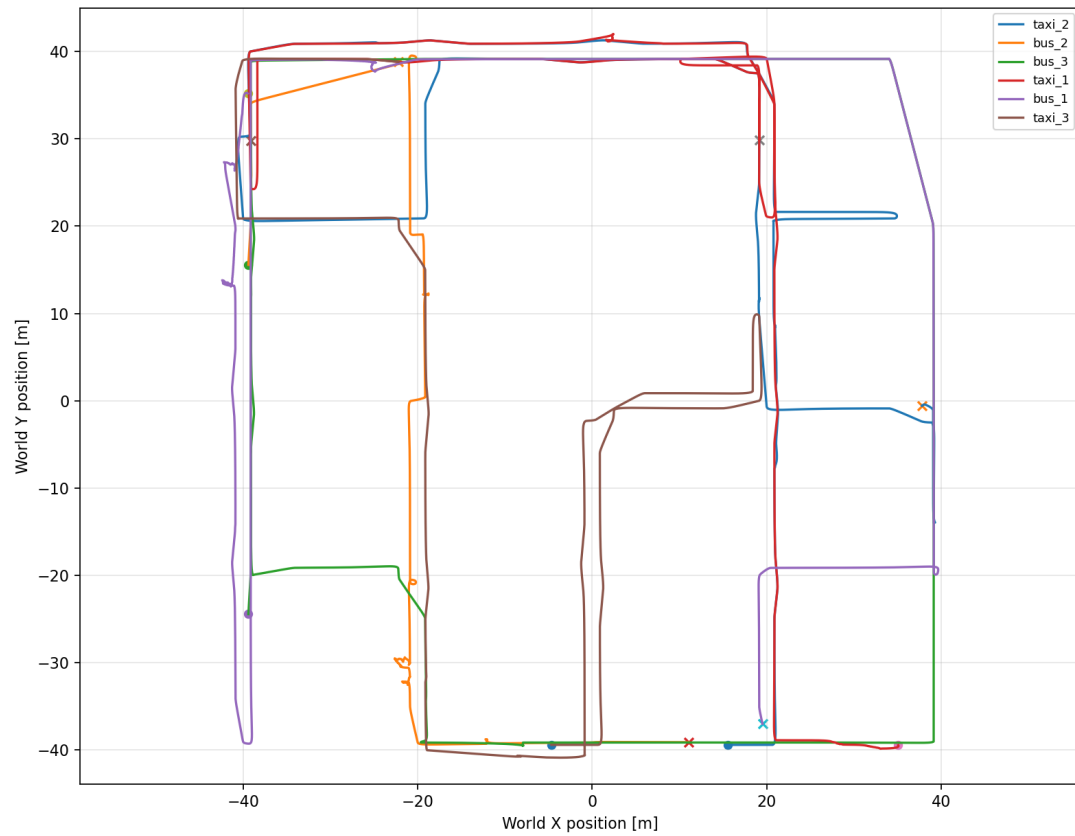


Figure 4.1: Vehicle trajectories recorded during the simulation. The plot shows the paths followed by buses and taxis within the urban road network, confirming that the vehicles were able to navigate through the city while sharing the same environment.

For each generated request, only one taxi was selected to perform the transportation task, preventing multiple assignments and conflicts between vehicles. The system therefore demonstrated that transportation requests can be managed without relying on a dedicated central coordination node.

4.4 Final Considerations

Overall, the simulation demonstrated how a network of autonomous agents can effectively operate within a shared urban environment.

The obtained results confirm the validity of the proposed architecture and demonstrate that ROS2 and Gazebo provide suitable tools for the development and experimentation of multi-agent systems intended for smart mobility scenarios.